



Alzikrayat Project Report

Web Advanced project [1]

- ايثار عماد الدين احمد
- تقانة معلومات – الفصل الدراسي السابع

1. Project Overview:

- Alzikrayat is a photo application for web advanced course project.
- The entire application is hand written in PHP using custom MVC (Model-View-Controller) framework combined with physical 3-Tier architecture. Without using any external backend frameworks, CMS, or ORM.
- Registered users can upload photos with filter or without, browse a shared community gallery and tag another users in his own photos, leave comments and have public profile page everyone can see it.
- Technologies: PHP 8 (procedural + object-oriented) – MySQL (PDO and prepared queries) – HTML5 – Bootstrap 5 (for a responsive interface, including its built-in dark/light color modes) – javascript (client-side validation, canvas-based photo filters, theme toggle) – css-regex (for manual router, validation)

2. Architecture & project structure:

- It follow the 3-tier architecture mapped onto the mvc pattern
- Routing is entirely manual. Routes registered in routes/web.php as method+path+[controller+action function] triples ,and core/Router.php convert path into a regEx pattern at dispatch time
- every income requests passes through the single front controller (public/index.php), which loads configuration, autoloads classes, starts the session, and hands the request to the router

- **Folder structure:**

config/ app.php (constants, paths) , database.php (PDO singleton connection)

core/ Router, base Model/Controller, Auth, Session, Request, Response, Validator, helpers

controllers/ AuthController, PhotoController, CommentController, HomeController, UserController

models/ User, Photo, Comment, Tag (one class per database table)

public/ the single front controller (index.php), CSS/JS assets, and the photo uploads folder+ App icon

views/ HTML/PHP templates, grouped by feature (auth, photos, home, users) plus a shared layout

databaseSQL/ schema.sql (the full database definition)

3. Database Design:

- alzikrayat database is normalized, with numeric auto-increments primary keys and foreign keys enforce referential integritys (cascading deletes)all queries wrote by hand with PDO prepared statements. Using index in every table
- Passwords never stored in plain text: password_hash() (Bcrypt) is used at registration, and password_verify() at login. Physical image files are stored on disk undern public/images/uploads/, while only the generated random file name and descriptive metadata are stored in the database

- **Tables:**

1- users

- registered accounts: name, email, hashed password, location, occupation, description
- relationship: referenced by photos, comments, photo_tags

2- photos

- One row for uploaded photo: file name, title, description, uploader, timestamp
- Relationship: user_id -> users.id (cascade delete)

3- comments

- One row for one comment on a photo
- relationships: photo_id -> photos.id, user_id -> users.id (cascade delete)

4- photo_tags

- Join table: which users are tagged in which photo
- photo_id -> photos.id, user_id -> users.id, unique pair (cascade delete)

4. Main Features:

- authentication: registration and login with server-side validation, Bcrypt password hashing, and session-based state
- "Last login" cookie: a separate 7-day browser cookie records the timestamp of the last successful login and displays it on the login page, independent of the session
- Photo management: upload (with file-type and size validation), a gallery with multiple display styles (3-column, 4-column, list, featured), a detail page, and owner-only deletion
- Comments: any logged-in user can comment on any photo; the comment list updates immediately after posting
- Three-level validation: HTML5 attributes, custom JavaScript, and server-side PHP validation for every form
- Security: PDO prepared statements everywhere (no string-concatenated SQL), output escaping with htmlspecialchars() against XSS, and CSRF tokens on every state-changing form
- Responsive UI: built entirely with Bootstrap 5, tested across mobile, tablet, and desktop breakpoints

5. Novelty/Advanced Features:

- **Tagging registered users in photos:** while uploading a photo (or afterwards, as the owner), a user can tag other registered users. Tagged names are shown on the photo's detail page and link to that person's profile — implemented with a dedicated photo_tags table and Tag model.
- **Canvas-based photo filters:** the upload page shows a live preview of the selected photo with Monochrome, Warm, and Soft-focus filters, implemented with the HTML5 Canvas pixel API. The chosen filter is re-applied to the photo at full resolution and replaces the original file just before it is uploaded, so the stored photo actually keeps the filter rather than only the on-screen preview.
- **Public profile pages:** every user has a profile page reachable from their name anywhere it appears — photo authorship, comments, or tags, showing their own uploaded photos and the photos they have been tagged in.
- **Dark / Light mode:** a toggle switches the whole interface between Bootstrap 5's built-in color modes. The preference is stored in a cookie and read on the server before the page is rendered, so the correct theme applies immediately with no flash of the wrong color scheme.

6. Conclusion:

Alzikrayat meets every requirement of the course specification: a hand-written MVC framework layered over a physical 3-Tier architecture, a manual regex-based router, raw parameterized SQL with no ORM, secure session-based authentication with a 7-day last-login cookie, full photo and comment CRUD with ownership checks, three-level input validation, and a responsive Bootstrap interface — with zero external backend frameworks used anywhere in the codebase.